

B6: Condensed-Matter Physics — 2024 Model Solutions

Trinity Term 2024 (Paper A13485W1)

Question 1 — Sodium under pressure: bcc–fcc transition and X-ray diffraction

Problem. Na is bcc at $p = 0$ ($a_0 = 4.27 \text{ \AA}$) and undergoes a first-order bcc→fcc transition at $p_c = 65 \text{ GPa}$, with $a = 2.90 \text{ \AA}$ just below and $a' = 3.65 \text{ \AA}$ just above. Powder XRD uses $\lambda = 0.43 \text{ \AA}$; molar mass $M = 23 \text{ g mol}^{-1}$.

(a)(i)–(ii) Conventional unit cells and atoms per cell



Count corners (each shared $\times 1/8$) plus interior atoms:

$$n_{\text{bcc}} = 8 \times \frac{1}{8} + 1 = 2.$$

$$n_{\text{fcc}} = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4.$$

$n_{\text{bcc}} = 2, \quad n_{\text{fcc}} = 4.$

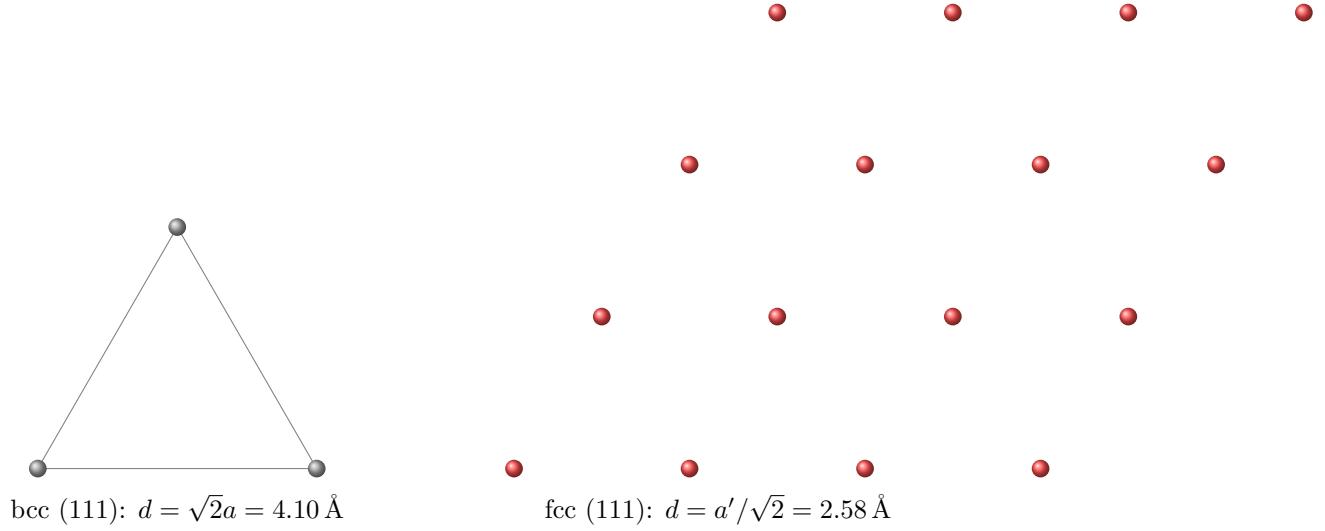
(a)(iii)–(iv) The (111) plane and nearest-neighbour distance in it

bcc (111) contains only the three corners (the body-centre atom is off the plane); side $= \sqrt{2}a$:

$$d_{\text{bcc}}^{(111)} = \sqrt{2} \times 2.90 \text{ \AA} = 4.10 \text{ \AA}.$$

fcc (111) is close-packed with separation $a'/\sqrt{2}$:

$$d_{\text{fcc}}^{(111)} = 3.65 \text{ \AA} / \sqrt{2} = 2.58 \text{ \AA}.$$



$$d_{\text{bcc}}^{(111)} = 4.10 \text{ Å}, \quad d_{\text{fcc}}^{(111)} = 2.58 \text{ Å}.$$

(b)(i) Diffraction selection rules and reflection positions

Bragg + cubic spacing $d_{hkl} = a/\sqrt{h^2 + k^2 + \ell^2}$:

$$\sin \theta = \frac{\lambda}{2a} \sqrt{N}, \quad N \equiv h^2 + k^2 + \ell^2. \quad (1)$$

Selection rules:

$$\boxed{\text{bcc: } h + k + \ell \text{ even; } \quad \text{fcc: } h, k, \ell \text{ all even or all odd.}} \quad (2)$$

Allowed N sequences:

$$N_{\text{bcc}} : 2, 4, 6, 8, \dots \quad (hkl) = 110, 200, 211, 220, \dots$$

$$N_{\text{fcc}} : 3, 4, 8, 11, 12, \dots \quad (hkl) = 111, 200, 220, 311, 222, \dots$$

Window $2\theta < 25^\circ$ requires $\sin \theta < 0.2164$.

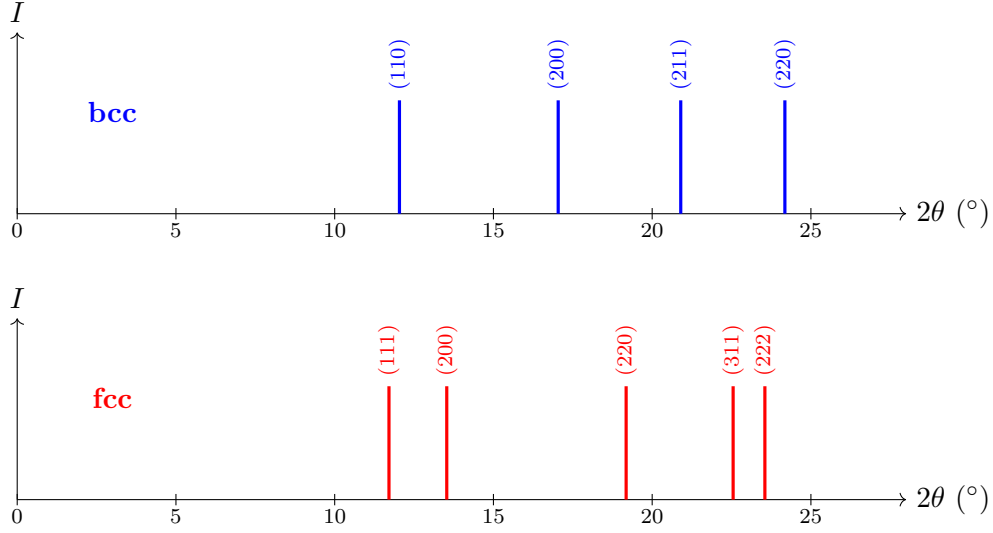
bcc, $\lambda/(2a) = 0.07414$, $\sqrt{N} < 2.919$, $N \leq 8$:

$$\begin{aligned} (110) : N = 2, \quad 2\theta &= 12.04^\circ; \\ (200) : N = 4, \quad 2\theta &= 17.04^\circ; \\ (211) : N = 6, \quad 2\theta &= 20.90^\circ; \\ (220) : N = 8, \quad 2\theta &= 24.18^\circ. \end{aligned} \quad (3)$$

fcc, $\lambda/(2a') = 0.05890$, $\sqrt{N} < 3.674$, $N \leq 12$:

$$\begin{aligned} (111) : N = 3, \quad 2\theta &= 11.71^\circ; \\ (200) : N = 4, \quad 2\theta &= 13.53^\circ; \\ (220) : N = 8, \quad 2\theta &= 19.18^\circ; \\ (311) : N = 11, \quad 2\theta &= 22.55^\circ; \\ (222) : N = 12, \quad 2\theta &= 23.55^\circ. \end{aligned} \quad (4)$$

(b)(ii) Sketch of the two diffraction patterns



(b)(iii) Lattice parameter at $p_1 = 100$ GPa

Third allowed fcc reflection (220) at $2\theta = 19.18^\circ$ shifts up by 1.20° :

$$2\theta_{220}^{(1)} = 19.18 + 1.20 = 20.38^\circ.$$

Take half:

$$\sin \theta_{220}^{(1)} = \sin(10.19^\circ) = 0.17681.$$

Apply (1) with $\sqrt{N} = 2\sqrt{2}$:

$$a'_1 = \frac{\lambda\sqrt{N}}{2 \sin \theta_{220}^{(1)}} = \frac{0.43 \times 2\sqrt{2}}{2 \times 0.17681} \text{ \AA}.$$

$$\boxed{a'_1 = 3.44 \text{ \AA}.} \quad (5)$$

(c) Hard-sphere atomic radius and density change at p_c

bcc touch along body diagonal $\sqrt{3}a = 4r$:

$$r_{\text{bcc}} = \frac{\sqrt{3}}{4} \times 2.90 \text{ \AA} = 1.256 \text{ \AA}.$$

fcc touch along face diagonal $\sqrt{2}a' = 4r$:

$$r_{\text{fcc}} = \frac{\sqrt{2}}{4} \times 3.65 \text{ \AA} = 1.291 \text{ \AA}.$$

Number density $n = n_{\text{cell}}/a^3$:

$$n_b = 2/(2.90)^3 \text{ \AA}^{-3} = 8.20 \times 10^{28} \text{ m}^{-3}.$$

$$n_f = 4/(3.65)^3 \text{ \AA}^{-3} = 8.23 \times 10^{28} \text{ m}^{-3}.$$

Mass density $\rho = nM/N_A$:

$$\rho_{\text{bcc}} = \frac{8.20 \times 10^{28} \times 23 \times 10^{-3}}{6.022 \times 10^{23}} \text{ kg m}^{-3} = 3133 \text{ kg m}^{-3}.$$

$$\rho_{\text{fcc}} = \frac{8.226 \times 10^{28} \times 23 \times 10^{-3}}{6.022 \times 10^{23}} \text{ kg m}^{-3} = 3142 \text{ kg m}^{-3}.$$

Fractional change:

$$\boxed{\Delta\rho/\rho_{\text{bcc}} = +0.31\%, \quad \Delta\rho \approx +9 \text{ kg m}^{-3}.} \quad (6)$$

(d) Compressibility of the bcc phase between ambient and p_c

Per-atom volume $v = a^3/2$. Treat $\beta = -(1/V)\partial V/\partial p$ constant; integrate:

$$V(p) = V_0 e^{-\beta p}.$$

Solve for β at $p = p_c$:

$$\beta = -\frac{1}{p_c} \ln \frac{V}{V_0} = -\frac{3}{p_c} \ln \frac{a}{a_0}.$$

Evaluate $\ln(2.90/4.27) = -0.3866$, $p_c = 65 \times 10^9 \text{ Pa}$:

$$\beta_{\text{bcc}} = \frac{3 \times 0.3866}{6.5 \times 10^{10}} \text{ Pa}^{-1}.$$

$$\boxed{\beta_{\text{bcc}} = 1.78 \times 10^{-11} \text{ Pa}^{-1}.} \quad (7)$$

(e) Mass of sodium that fits in the pressure cell

Cylindrical volume:

$$V_{\text{cyl}} = \pi(50 \times 10^{-6})^2 \times 10^{-3} \text{ m}^3 = 7.854 \times 10^{-12} \text{ m}^3.$$

35% void $\Rightarrow$ Na occupies $0.65 V_{\text{cyl}}$:

$$V_{\text{Na}} = 0.65 \times 7.854 \times 10^{-12} \text{ m}^3 = 5.10 \times 10^{-12} \text{ m}^3.$$

Ambient bcc density with $a_0^3 = 77.85 \times 10^{-30} \text{ m}^3$:

$$\rho_0 = \frac{2 \times 23 \times 10^{-3}}{6.022 \times 10^{23} \times 77.85 \times 10^{-30}} \text{ kg m}^{-3} = 981 \text{ kg m}^{-3}.$$

Mass:

$$m_{\text{Na}} = 981 \times 5.10 \times 10^{-12} \text{ kg}.$$

$$\boxed{m_{\text{Na}} \approx 5.0 \times 10^{-9} \text{ kg} \approx 5 \text{ ng}.} \quad (8)$$

Question 2 — Heat capacity of YbMg₂Bi₂ in 3D and 2D

Problem. 3D molar heat capacity is $C = zN_A k_B [(\pi^2/2)(T/T_F) + (12\pi^4/5)(T/T_D)^3]$ at low T , saturating at high T . (a) Identify z from the saturation. (b) Justify each term, extract T_F, T_D from C/T vs T^2 , find cross-over T^* . (c) For a 2D film with two in-plane polarisations, find $g_{\text{ph}}^{(2)}(\omega)$, the high- T limit, and the low- T form $C = AT + BT^\delta$ with $A(g(E_F))$, $B(\omega_D)$. Use $\int_0^\infty x^3/(e^x - 1) dx = \pi^4/15$, $\int_0^\infty x^2/(e^x - 1) dx = 2\zeta(3) \simeq 2.4$.

(a) Origin of the high- T saturation and Dulong–Petit limit

Equipartition for $3N$ classical modes:

$$C_{\text{lat}}^{(\infty)} = 3Nk_{\text{B}}.$$

Per mole of formula units, $N = zN_{\text{A}}$:

$$C^{(\infty)} = 3zR.$$

Read $C^{(\infty)} \simeq 120 \text{ J mol}^{-1} \text{ K}^{-1}$ from figure:

$$z = 120/(3 \times 8.314) \approx 4.8.$$

$$\boxed{z = 5 \text{ atoms per formula unit.}} \quad (9)$$

(b)(i) Justification of the two terms

Linear electronic term. Sommerfeld expansion to leading order in T/T_{F} :

$$U_{\text{el}}(T) = U_{\text{el}}(0) + \frac{\pi^2}{6} g(E_{\text{F}}) (k_{\text{B}}T)^2. \quad (10)$$

Differentiate:

$$C_{\text{el}} = \frac{\pi^2}{3} g(E_{\text{F}}) k_{\text{B}}^2 T \equiv \gamma T. \quad (11)$$

Free-electron-like: $g(E_{\text{F}}) = 3N_{\text{e}}/(2k_{\text{B}}T_{\text{F}})$. Substitute:

$$C_{\text{el}} = \frac{\pi^2}{3} \cdot \frac{3N_{\text{e}}}{2k_{\text{B}}T_{\text{F}}} k_{\text{B}}^2 T = \frac{\pi^2}{2} N_{\text{e}} k_{\text{B}} \frac{T}{T_{\text{F}}}.$$

Identify $N_{\text{e}} = zN_{\text{A}}$ to recover the stated prefactor.

Cubic Debye term. 3D phonon DOS (3 acoustic branches):

$$g_{\text{ph}}(\omega) = \frac{3V\omega^2}{2\pi^2 v_{\text{s}}^3}. \quad (12)$$

Internal energy:

$$U_{\text{ph}}(T) = \int_0^{\omega_{\text{D}}} \frac{\hbar\omega}{e^{\hbar\omega/k_{\text{B}}T} - 1} g_{\text{ph}}(\omega) d\omega.$$

Substitute $x = \hbar\omega/k_{\text{B}}T$:

$$U_{\text{ph}}(T) = \frac{3V}{2\pi^2 v_{\text{s}}^3 \hbar^3} (k_{\text{B}}T)^4 \int_0^{x_{\text{D}}} \frac{x^3}{e^x - 1} dx.$$

Low T , extend the upper limit to ∞ :

$$U_{\text{ph}}(T) \approx \frac{3V}{2\pi^2 v_{\text{s}}^3 \hbar^3} (k_{\text{B}}T)^4 \cdot \frac{\pi^4}{15}.$$

Eliminate V/v_{s}^3 via $\omega_{\text{D}}^3 = 6\pi^2 N v_{\text{s}}^3/V$:

$$\frac{V}{v_{\text{s}}^3} = \frac{6\pi^2 N \hbar^3}{(k_{\text{B}}T_{\text{D}})^3}.$$

Substitute:

$$U_{\text{ph}}(T) = \frac{3\pi^4}{5} N k_{\text{B}} \frac{T^4}{T_{\text{D}}^3}.$$

Differentiate, use $N = zN_{\text{A}}$:

$$\boxed{C_{\text{ph}}(T) = \frac{12\pi^4}{5} z N_{\text{A}} k_{\text{B}} \frac{T^3}{T_{\text{D}}^3}.} \quad (13)$$

(b)(ii) Reading T_F and T_D from the C/T vs T^2 plot

Model:

$$\frac{C}{T} = \gamma + \beta T^2, \quad \gamma = \frac{\pi^2 z R}{2T_F}, \quad \beta = \frac{12\pi^4 z R}{5T_D^3}. \quad (14)$$

Read intercept $\gamma \simeq 0.30 \text{ J mol}^{-1} \text{ K}^{-2}$, slope $\beta \simeq 1.6 \times 10^{-3} \text{ J mol}^{-1} \text{ K}^{-4}$.

Solve for T_F :

$$T_F = \frac{\pi^2 \times 5 \times 8.314}{2 \times 0.30} \text{ K} \approx 684 \text{ K}.$$

$T_F \approx 7 \times 10^2 \text{ K}.$

(15)

Solve for T_D :

$$T_D^3 = \frac{12\pi^4 \times 5 \times 8.314}{5 \times 1.6 \times 10^{-3}} \text{ K}^3 = 6.07 \times 10^5 \text{ K}^3.$$

Cube root:

$$T_D \approx 195 \text{ K}.$$

$T_D \approx 2 \times 10^2 \text{ K}.$

(16)

(b)(iii) Cross-over temperature T^*

Equate the two contributions:

$$\frac{\pi^2}{2} \frac{T^*}{T_F} = \frac{12\pi^4}{5} \left(\frac{T^*}{T_D} \right)^3.$$

Solve for $(T^*)^2$:

$$(T^*)^2 = \frac{5T_D^3}{24\pi^2 T_F}.$$

Plug numbers:

$$(T^*)^2 = \frac{5 \times 195^3}{24\pi^2 \times 684} \text{ K}^2 = 229 \text{ K}^2.$$

$T^* \approx 15 \text{ K}.$

(17)

(c)(i) 2D phonon density of states

Two in-plane polarisations, mode density $A/(2\pi)^2$ per polarisation, $\omega = v_s |\mathbf{q}|$:

$$g_{\text{ph}}^{(2)}(\omega) d\omega = 2 \cdot \frac{A}{(2\pi)^2} \cdot 2\pi q dq.$$

Insert $q = \omega/v_s$:

$$g_{\text{ph}}^{(2)}(\omega) d\omega = \frac{A\omega}{\pi v_s^2} d\omega.$$

$g_{\text{ph}}^{(2)}(\omega) = \frac{A\omega}{\pi v_s^2}.$

(18)

(c)(ii) High- T Dulong–Petit in 2D

Total in-plane modes = $2N$, each k_B :

$$C_{2D}^{(\infty)} = 2Nk_B.$$

Per mole, $N = zN_A$:

$$C_{\text{molar}}^{(\infty)} = 2zR = 10R \approx 83 \text{ J mol}^{-1} \text{ K}^{-1}. \quad (19)$$

(c)(iii) Low- T form $C = AT + BT^\delta$ in 2D

Phonon energy with $g_{\text{ph}}^{(2)}$:

$$U_{\text{ph}}^{(2)}(T) = \int_0^{\omega_D} \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1} \frac{A\omega}{\pi v_s^2} d\omega.$$

Substitute $x = \hbar\omega/k_B T$:

$$U_{\text{ph}}^{(2)}(T) = \frac{A}{\pi v_s^2 \hbar^2} (k_B T)^3 \int_0^{x_D} \frac{x^2}{e^x - 1} dx.$$

Low T , extend the upper limit to ∞ :

$$U_{\text{ph}}^{(2)}(T) \approx \frac{A}{\pi v_s^2 \hbar^2} (k_B T)^3 \cdot 2\zeta(3).$$

Differentiate:

$$C_{\text{ph}}^{(2)}(T) = \frac{6\zeta(3)Ak_B^3}{\pi v_s^2 \hbar^2} T^2 \equiv BT^\delta, \quad \delta = 2. \quad (20)$$

(c)(iv) Identification of A in terms of $g(E_F)$, and B in terms of ω_D

Sommerfeld result (11) is dimension-independent:

$$A = \frac{\pi^2}{3} k_B^2 g(E_F). \quad (21)$$

Debye condition (total in-plane modes = $2N$):

$$2N = \int_0^{\omega_D} \frac{A\omega}{\pi v_s^2} d\omega = \frac{A\omega_D^2}{2\pi v_s^2}.$$

Rearrange:

$$\frac{A}{\pi v_s^2} = \frac{4N}{\omega_D^2}.$$

Substitute into (20):

$$C_{\text{ph}}^{(2)} = \frac{6\zeta(3)k_B^3}{\hbar^2} \cdot \frac{4NT^2}{\omega_D^2} = \frac{24\zeta(3)Nk_B^3}{\hbar^2 \omega_D^2} T^2.$$

Per mole, $N = zN_A$:

$$B = \frac{24\zeta(3)zN_A k_B^3}{\hbar^2 \omega_D^2}. \quad (22)$$

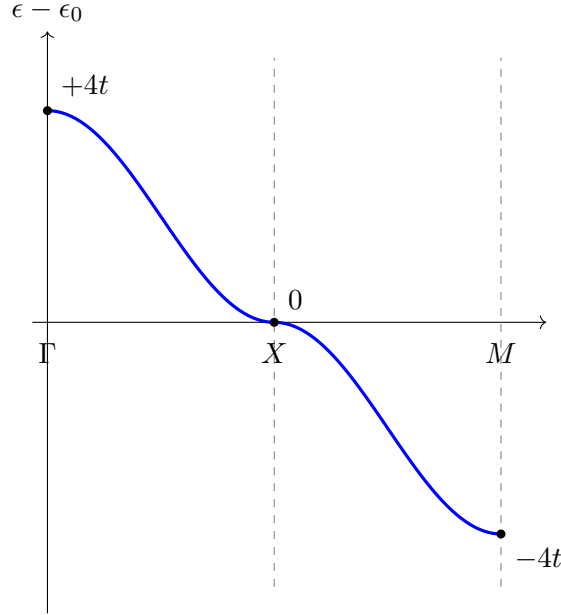
Question 3 — Tight-binding band on a 2D square lattice

Problem. 2D square lattice (parameter a), tight-binding $\epsilon(k_x, k_y) = \epsilon_0 + 2t \cos k_x a + 2t \cos k_y a$, $t > 0$. Find dispersion along $\Gamma \rightarrow X \rightarrow M$, bandwidth, contours, effective masses; the linear $\epsilon_F = \epsilon'_F + \alpha N/A$ near the band minimum and $g(\epsilon) = \beta A$; for $A_{FS}/A_{BZ} = 1/100$ at $a = 3.5 \text{ \AA}$, find n_{2D} and Hall voltage.

(a)(i) Band dispersion along $\Gamma \rightarrow X \rightarrow M$

Endpoint values:

$$\epsilon(\Gamma) = \epsilon_0 + 4t, \quad \epsilon(X) = \epsilon_0, \quad \epsilon(M) = \epsilon_0 - 4t.$$



(a)(ii) Bandwidth

$$W = \epsilon_{\max} - \epsilon_{\min} = 8t. \quad (23)$$

(a)(iii) Constant-energy contours

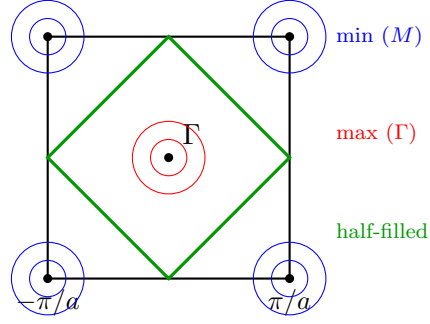
Expand around $M = (\pi/a, \pi/a)$, $\mathbf{k} = \mathbf{K}_M + \mathbf{q}$, $\cos(\pi + qa) \simeq -1 + (qa)^2/2$:

$$\epsilon(\mathbf{k}) \simeq \epsilon_0 - 4t + ta^2(q_x^2 + q_y^2). \quad (24)$$

Expand around Γ , $\cos(qa) \simeq 1 - (qa)^2/2$:

$$\epsilon(\mathbf{k}) \simeq \epsilon_0 + 4t - ta^2(q_x^2 + q_y^2). \quad (25)$$

Both circular. Half-filling $\epsilon = \epsilon_0$: $\cos k_x a = -\cos k_y a \Rightarrow k_x a = \pm(\pi - k_y a)$, a square rotated 45° .



(a)(iv) Effective masses near band edges

From (24), $\partial^2 \epsilon / \partial q^2 = 2ta^2$:

$$m_{\min}^* = \frac{\hbar^2}{2ta^2} > 0.$$

From (25), $\partial^2 \epsilon / \partial q^2 = -2ta^2$:

$$m_{\max}^* = -\frac{\hbar^2}{2ta^2} < 0.$$

$$\boxed{m_{\min}^* = +\frac{\hbar^2}{2ta^2}, \quad m_{\max}^* = -\frac{\hbar^2}{2ta^2}. \quad (26)}$$

(b)(i)–(ii) $\epsilon_F(N/A)$ near the band minimum

Parabolic band, $m^* = \hbar^2/(2ta^2)$. Count $\mathbf{k}$ -states inside contour, factor 2 for spin:

$$N(\epsilon_F) = 2 \cdot \frac{A}{(2\pi)^2} \cdot \pi q_F^2.$$

From (24): $ta^2 q_F^2 = \epsilon_F - (\epsilon_0 - 4t)$. Substitute:

$$\frac{N}{A} = \frac{q_F^2}{2\pi} = \frac{\epsilon_F - \epsilon_0 + 4t}{2\pi ta^2}.$$

Solve for ϵ_F :

$$\boxed{\epsilon_F = (\epsilon_0 - 4t) + 2\pi ta^2 \frac{N}{A}, \quad \epsilon'_F = \epsilon_0 - 4t, \quad \alpha = 2\pi ta^2. \quad (27)}$$

(b)(iii) Density of states near the band minimum

Differentiate N with respect to ϵ_F :

$$g(\epsilon) = \frac{\partial N}{\partial \epsilon} = \frac{A}{2\pi ta^2}.$$

$$\boxed{\beta = \frac{1}{2\pi ta^2}. \quad (28)}$$

(c)(i) **Carrier density for** $A_{\text{FS}}/A_{\text{BZ}} = 1/100$

BZ area $A_{\text{BZ}} = (2\pi/a)^2$. Spin factor 2 over $\mathbf{k}$ -cell:

$$\frac{N}{A} = 2 \cdot \frac{A_{\text{FS}}}{(2\pi)^2} = \frac{2}{100} \cdot \frac{(2\pi/a)^2}{(2\pi)^2}.$$

Simplify:

$$\frac{N}{A} = \frac{1}{50a^2}.$$

For $a = 3.5 \times 10^{-10}$ m:

$$n_{2D} = \frac{1}{50 \times (3.5 \times 10^{-10})^2} \text{ m}^{-2}.$$

$$\boxed{n_{2D} = 1.63 \times 10^{17} \text{ m}^{-2}.} \quad (29)$$

(c)(ii) **Hall voltage**

Hall coefficient (single carrier, magnitude):

$$R_H^{(2D)} = \frac{1}{n_{2D} e}.$$

Hall field per unit width:

$$V_H = \frac{j_x B_z}{n_{2D} e}.$$

Plug $j_x = 1.0 \times 10^{-5} \text{ A m}^{-1}$, $B_z = 1 \text{ T}$:

$$V_H = \frac{1.0 \times 10^{-5} \times 1}{1.63 \times 10^{17} \times 1.602 \times 10^{-19}} \text{ V}.$$

Evaluate denominator:

$$1.63 \times 10^{17} \times 1.602 \times 10^{-19} = 2.61 \times 10^{-2}.$$

Final:

$$\boxed{V_H \approx 3.8 \times 10^{-4} \text{ V} \approx 0.38 \text{ mV}.} \quad (30)$$

Question 4 — Magnetism: Hund's rules, mean-field BCC ferromagnet

Problem. (a) Hund's rules for Mn^{2+} ($3d^5$), Tb^{3+} ($4f^8$); $T = 0$ moments. (b) $S = 1/2$ Heisenberg + Zeeman on bcc (edge a): mean-field B_{mf} , self-consistent M , Curie-Weiss χ , $\Theta = T_C$, find J_1 for $T_C = 30 \text{ K}$. (c) Same on simple cubic with $T'_C = 20 \text{ K}$; find J'_1 ; sketch $\chi^{-1}(T)$.

(a)(i) **Hund's rules for Mn^{2+} ($3d^5$) and Tb^{3+} ($4f^8$)**

Mn^{2+} , $3d^5$. Five d electrons, one in each $m_\ell \in \{-2, \dots, 2\}$:

$$S = 5 \times \frac{1}{2} = \frac{5}{2}, \quad L = \sum m_\ell = 0.$$

Half-filled, $J = L + S$ or $|L - S|$ both give:

$$J = \frac{5}{2}.$$

Landé factor:

$$g_J = \frac{3}{2} + \frac{(5/2)(7/2) - 0}{2(5/2)(7/2)} = 2.$$

$$\boxed{\text{Mn}^{2+} : S = \frac{5}{2}, L = 0, J = \frac{5}{2}, g_J = 2.} \quad (31)$$

Tb³⁺, 4f⁸. Seven aligned, eighth pairs into $m_\ell = +3$:

$$S = \frac{7}{2} - \frac{1}{2} = 3, \quad L = 0 + 3 = 3.$$

More than half full, $J = L + S$:

$$J = 6.$$

Landé factor:

$$g_J = \frac{3}{2} + \frac{12 - 12}{2 \cdot 6 \cdot 7} = \frac{3}{2}.$$

$$\boxed{\text{Tb}^{3+} : S = 3, L = 3, J = 6, g_J = \frac{3}{2}.} \quad (32)$$

(a)(ii) Saturation magnetic moment at $T = 0$

$m = g_J J \mu_B$:

$$m_{\text{Mn}^{2+}} = 2 \times \frac{5}{2} \mu_B = 5 \mu_B.$$

$$m_{\text{Tb}^{3+}} = \frac{3}{2} \times 6 \mu_B = 9 \mu_B.$$

$$\boxed{m_{\text{Mn}^{2+}} = 5 \mu_B, \quad m_{\text{Tb}^{3+}} = 9 \mu_B.} \quad (33)$$

(b)(i) Mean-field effective field on a bcc site

bcc coordination $z = 8$. Replace $\mathbf{S}_j \rightarrow \langle \mathbf{S} \rangle$ in the Heisenberg sum:

$$\mathcal{H}_i^{\text{MF}} = -J_1 z \langle \mathbf{S} \rangle \cdot \mathbf{S}_i + g \mu_B \mathbf{B} \cdot \mathbf{S}_i \equiv g \mu_B \mathbf{B}_{\text{mf}} \cdot \mathbf{S}_i. \quad (34)$$

Read off:

$$\mathbf{B}_{\text{mf}} = \mathbf{B} - \frac{J_1 z}{g \mu_B} \langle \mathbf{S} \rangle.$$

Use $\mathbf{M} = -n g \mu_B \langle \mathbf{S} \rangle$, $n = 2/a^3$:

$$\langle \mathbf{S} \rangle = -\frac{\mathbf{M}}{n g \mu_B}.$$

Substitute:

$$\mathbf{B}_{\text{mf}} = \mathbf{B} + \frac{J_1 z}{(g \mu_B)^2 n} \mathbf{M} \equiv \mathbf{B} + \lambda \mathbf{M}.$$

Insert $z = 8$, $n = 2/a^3$:

$$\boxed{\lambda = \frac{4 J_1 a^3}{(g \mu_B)^2}.} \quad (35)$$

(b)(ii) Self-consistent magnetisation

Single $S = 1/2$ in B_{mf} :

$$Z = 2 \cosh\left(\frac{g\mu_B B_{\text{mf}}}{2k_B T}\right).$$

Average projection $\frac{1}{2} \tanh(\cdot)$, multiply by moment density $ng\mu_B$:

$$M = \frac{ng\mu_B}{2} \tanh\left(\frac{g\mu_B B_{\text{mf}}}{2k_B T}\right) \equiv M_0 \tanh\left(\frac{g\mu_B B_{\text{mf}}}{2k_B T}\right).$$

Insert $n = 2/a^3$:

$$\boxed{M_0 = \frac{g\mu_B}{a^3}.} \quad (36)$$

(b)(iii) Curie–Weiss susceptibility above T_C

Expand $\tanh x \approx x$ for small M :

$$M \simeq M_0 \cdot \frac{g\mu_B (B + \lambda M)}{2k_B T}.$$

Collect M :

$$M \left(1 - \frac{M_0 g\mu_B \lambda}{2k_B T}\right) = \frac{M_0 g\mu_B B}{2k_B T}.$$

Identify $\chi = M/B$:

$$\chi = \frac{M_0 g\mu_B / (2k_B)}{T - M_0 g\mu_B \lambda / (2k_B)} \equiv \frac{C}{T - \Theta}.$$

Substitute M_0, λ from (36),(35):

$$\Theta = \frac{g\mu_B}{a^3} \cdot g\mu_B \cdot \frac{4J_1 a^3}{(g\mu_B)^2} \cdot \frac{1}{2k_B} = \frac{2J_1}{k_B}.$$

$$C = \frac{(g\mu_B)^2}{2k_B a^3}.$$

$$\boxed{\Theta = \frac{2J_1}{k_B}, \quad C = \frac{(g\mu_B)^2}{2k_B a^3}.} \quad (37)$$

(b)(iv) Relation $\Theta = T_C$

$B = 0$ self-consistent equation $M = M_0 \tanh(g\mu_B \lambda M / (2k_B T))$. Non-zero solution emerges when slope at $M = 0$ equals one:

$$\frac{M_0 g\mu_B \lambda}{2k_B T_C} = 1.$$

Identical to the pole condition in χ :

$$\boxed{T_C = \Theta = \frac{2J_1}{k_B}.} \quad (38)$$

(b)(v) Numerical J_1 for $T_C = 30$ K

$$J_1 = \frac{k_B T_C}{2} = \frac{1.381 \times 10^{-23} \times 30}{2} \text{ J}.$$

$$\boxed{J_1 \approx 2.07 \times 10^{-22} \text{ J} \approx 1.3 \text{ meV}, \quad J_1/k_B \approx 15 \text{ K}.} \quad (39)$$

(c)(i) J'_1 for the simple-cubic lattice with $T'_C = 20$ K

Simple cubic: $z' = 6$, $n' = 1/a^3$. Repeat (b)(i)–(iii):

$$\lambda' = \frac{6J'_1 a^3}{(g\mu_B)^2}, \quad M'_0 = \frac{g\mu_B}{2a^3}.$$

Compute Θ' :

$$\Theta' = \frac{M'_0 g\mu_B \lambda'}{2k_B} = \frac{3J'_1}{2k_B}.$$

Set $T'_C = \Theta'$ and solve:

$$J'_1 = \frac{2k_B T'_C}{3} = \frac{2 \times 1.381 \times 10^{-23} \times 20}{3} \text{ J}.$$

$$\boxed{J'_1 \approx 1.84 \times 10^{-22} \text{ J} \approx 1.15 \text{ meV}, \quad J'_1/k_B \approx 13.3 \text{ K}.} \quad (40)$$

(c)(ii) Sketch of $\chi^{-1}(T)$ for both materials

$\chi^{-1} = (T - \Theta)/C$, slope $1/C$, T -intercept Θ . With $C_{\text{bcc}} = (g\mu_B)^2/(2k_B a^3)$ and $C_{\text{sc}} = (g\mu_B)^2/(4k_B a^3)$, the sc slope is twice the bcc slope.

